R for Data Analysis & Visualization Final Project: Statistics Dashboard with Quarto

SI Analytics

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Introduction

Welcome to the final project of the **R for Data Analysis and Visualization** course.

For this project, you will build a professional, Quarto report that explores and visualises real-world data from Gapminder. This project is designed to help you consolidate your skills in data import, cleaning, transformation, visualisation, dashboard design, and analytical storytelling using R.

You can view the sample report by clicking on the link below.

VIEW SAMPLE REPORT IN NEW TAB

Your final quarto report should be portfolio-ready and suitable for sharing with future employers, collaborators, or professional networks.

Systematic vs just-in-time learning

As you work on this project, you will encounter many functions and concepts you have not previously learned. You'll therefore be required to pick these up "on the go" as you move through the analysis. This type of just-in-time learning, where you acquire knowledge exactly at the point when you need it, is quite different from the systematic, front-loaded learning approach that we have used throughout the course thus far, where all relevant information was presented before it was applied or tested.

Just-in-time learning will require you to be proactive in seeking out resources and asking for help when needed. It may be challenging at times, but it is a valuable skill to have as a programmer and will serve you well in your future career.

Project Requirements

Data Selection

- Choose at least **two indicators** from the Gapminder data repository.
- You may subset your analysis to a single year or analyse patterns across multiple years.
- Your two indicators should allow for meaningful descriptive comparison or relationship exploration. Examples include:
  - Life expectancy and income per person
  - Child mortality and health expenditure
  - Population and CO_2 emissions
  - Fertility rate and child mortality
  - Literacy rate and income per person
- If you would prefer to use alternative datasets that are more relevant to your professional work, please prepare to discuss your proposal with the instructional team during the week 8 live workshop. Your final dashboard must still meet the requirements in the grading rubric.

Report Structure

Your report should contain **at least three pages**, including the following:

1. **First Indicator Analysis**
   Explore trends, distributions, rankings, or geographic patterns for your first selected indicator.
2. **Second Indicator Analysis**
   Explore trends, distributions, rankings, or geographic patterns for your second selected indicator.
3. **Relationship Between the Two Indicators**
   Compare both indicators descriptively using charts, tables, or other dashboard elements. No formal statistical modelling is required. The relationship analysis may be purely descriptive and graphical.

Technical Requirements

Your dashboard should include at least **12 display elements**.

Display elements may include:

- Value boxes
- Interactive plots
- Static plots
- Tables
- Maps
- Text boxes with descriptions of key insights
- Summary cards
- Trend charts
- Country comparison charts
- Distribution plots

You should demonstrate your ability to use a range of R data manipulation and visualisation tools, including:

- `dplyr` for data wrangling
- `tidyr` for reshaping data
- `readr` or related packages for data import
- `ggplot2` for visualisation
- `plotly` for interactive charts
- `DT`, `reactable`, or similar packages for interactive tables
- Quarto dashboard features for layout and presentation

Your code should show evidence of substantial data preparation, including:

- Cleaning column names
- Selecting and renaming variables
- Filtering rows
- Creating new variables
- Grouping and summarising data
- Joining datasets when needed
- Pivoting data with `pivot_longer()` or `pivot_wider()`

Submission

You will be asked to submit:

- Your completed Quarto `.qmd` file
- Any supporting data files used
- A link to your deployed dashboard, preferably through GitHub Pages or another approved platform

Your report should render successfully before submission.

Grading Rubric

Total: **100 points**

Data Selection and Manipulation

25 points

Excellent submissions will:

- Select relevant indicators from Gapminder or an approved alternative source
- Clearly explain why the selected indicators are meaningful
- Import data correctly
- Clean data without errors or obvious inconsistencies
- Merge or join datasets correctly when needed

- Demonstrate strong use of R data wrangling functions from `dplyr` and `tidyr`
- Prepare a clean analytical dataset suitable for visualisation and dashboard reporting

Dashboard Structure

15 points

Excellent submissions will:

- Include the required three pages: first indicator, second indicator, and relationship analysis
- Use a logical dashboard layout
- Make effective use of space
- Organise content in a way that supports clear interpretation
- Use design choices that improve readability and understanding

Visualisation Elements

30 points

Excellent submissions will:

- Include at least 12 display elements
- Use appropriate visualisation types for the selected indicators
- Include clear chart titles, axis labels, legends, and scales
- Use interactive features effectively where appropriate
- Ensure hover text, filters, or tooltips are clear and useful
- Include text boxes or annotations that explain key insights
- Avoid cluttered or misleading visual design

Code Quality

10 points

Excellent submissions will:

- Use readable and well-organised code
- Include clear code chunk labels
- Run without errors
- Use efficient R functions and workflows
- Follow consistent formatting and style
- Avoid unnecessary or duplicated code

Deployment and Repository Structure

10 points

Excellent submissions will:

- Successfully deploy the dashboard to GitHub Pages or another approved platform
- Organise files logically in the project repository
- Include only necessary files
- Use clean and professional file names
- Ensure the deployed dashboard is accessible and renders correctly

Originality, Creativity, and Insight

10 points

Excellent submissions will:

- Demonstrate thoughtful exploration of the selected indicators
- Present interesting patterns, trends, or relationships
- Use creative but appropriate dashboard design choices
- Communicate insights clearly and professionally
- Show evidence of independent analytical thinking

Final Reminder

Your goal is not only to complete a technical exercise. Your goal is to produce a polished, professional dashboard that demonstrates your ability to use R and Quarto to turn real-world data into clear, meaningful, and shareable insights.